COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	9/17/2026	**DATE SUBMITTED:**	9/17/2026
NAME:	Momo, Borge Rudolf A.	**INSTRUCTOR:**	Engr. Quejado

ACTIVITY NO. 8.1 **Install, Configure, and Manage Enterprise Availability Monitoring via Ansible**

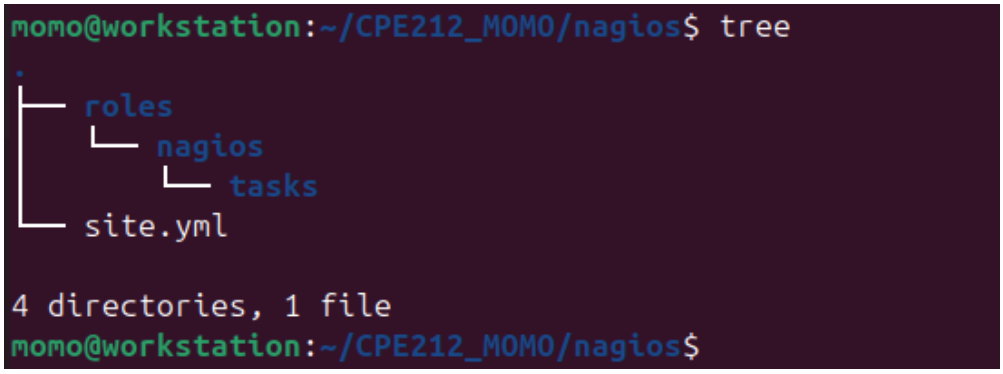
1. PROCEDURE OUTPUT

1. Create a playbook that installs Nagios in both Ubuntu and CentOS. Apply the concept of creating roles.

- What I did is create a whole playbook about installing Nagios in both Ubuntu and CentOS along with the basic configurations so that Nagios will work properly. The concept of creating roles was also implemented which the playbook will be debunked in number 2.

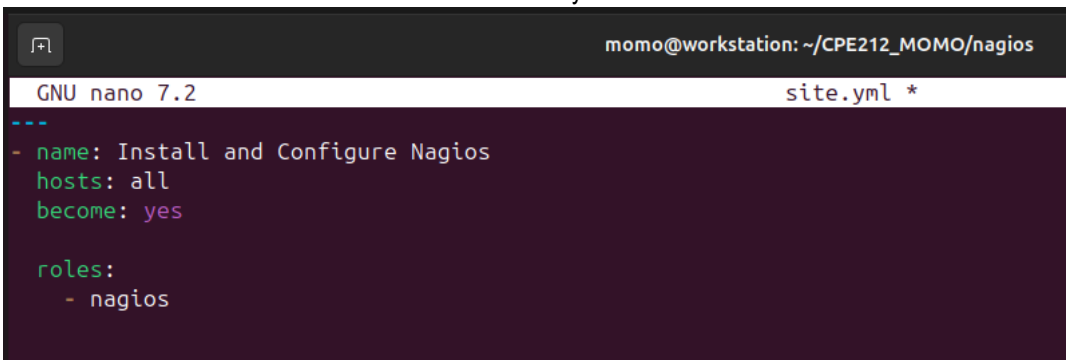
2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)

The Tree



- It shows the folders and files used for the Nagios Ansible playbook and it also shows that the concept of roles were applied

The site.yml



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- It is the main playbook that runs the Nagios role on the servers.

Install Nagios dependencies on Ubuntu

```
- name: Install Nagios dependencies on Ubuntu
  ansible.builtin.apt:
    name:
      - autoconf
      - gcc
      - libc6
      - make
      - wget
      - unzip
      - apache2
      - apache2-utils
      - php
      - libapache2-mod-php
      - libgd-dev
      - ufw
      - openssl
      - libssl-dev
      - gettext
      - automake
      - build-essential
      - snmp
      - libnet-snmp-perl
      - iputils-ping
    state: present
  when: ansible_distribution == "Ubuntu"
```

- This installs the required packages on the Ubuntu servers so Nagios can be built and run properly.

Install Nagios dependencies on CentOS

```
- name: same thing but for centos
  ansible.builtin.dnf:
    name:
      - gcc
      - glibc
      - glibc-common
      - perl
      - make
      - gettext
      - automake
      - autoconf
      - wget
      - gd
      - gd-devel
      - httpd
      - php
      - openssl-devel
      - net-snmp
      - net-snmp-utils
    state: present
  when: ansible_distribution == "CentOS"
```

- Also installs the required packages but for CentOS.

Create Nagios group

```
- name: Create Nagios group
  ansible.builtin.group:
    name: nagios
    state: present
```

- This creates a Nagios group that is used to manage the permissions of Nagios files.
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Create Nagios user

```
- name: Create Nagios user
  ansible.builtin.user:
    name: nagios
    group: nagios
    shell: /bin/bash
    state: present
```

- This creates a Nagios user that will be used to run and manage Nagios.

Add web server user to Nagios group

```
- name: Add web server user to Nagios group
  ansible.builtin.user:
    name: "{{ 'www-data' if ansible_distribution == 'Ubuntu' else 'apache' }}"
    groups: nagios
    append: yes
```

- This adds the web server user to the Nagios group so it can access the needed Nagios files.

Download Nagios Core

```
- name: Download Nagios Core
  ansible.builtin.get_url:
    url: https://github.com/NagiosEnterprises/nagioscore/releases/download/nagios-4.5.10/nagios-4.5.10.tar.gz
    dest: /tmp/nagioscore.tar.gz
```

- This just downloads the Nagios Core source files to the servers.

Extract Nagios Core

```
- name: Extract Nagios Core
  ansible.builtin.unarchive:
    src: /tmp/nagioscore.tar.gz
    dest: /tmp
    remote_src: yes
```

- This extracts the downloaded Nagios Core files so they can be configured and installed.

Configure Nagios Core

```
- name: Configure Nagios Core
  ansible.builtin.command: ./configure
  args:
    chdir: /tmp/nagios-4.5.10
```

- This configures Nagios Core based on the server setup before building it.

Build Nagios Core

```
- name: Build Nagios Core
  ansible.builtin.command: make all
  args:
    chdir: /tmp/nagios-4.5.10
```

- This compiles the Nagios Core source files so they can be installed on the servers.

Install Nagios Core

```
- name: Install Nagios service
  ansible.builtin.command: make install-daemoninit
  args:
    chdir: /tmp/nagios-4.5.10
```

- This installs the built Nagios Core files into the proper locations on the servers.

Enable Ubuntu Apache modules

```
- name: Enable Ubuntu Apache modules
  ansible.builtin.command: "{{ item }}"
  loop:
    - a2enmod rewrite
    - a2enmod cgi
  when: ansible_distribution == "Ubuntu"
```

- This enables the required Apache modules on Ubuntu so the Nagios web interface can work.

Download Nagios Plugins

```
- name: Download Nagios Plugins
  ansible.builtin.get_url:
    url: https://github.com/nagios-plugins/nagios-plugins/releases/download/release-2.4.11/nagios-plugins-2.4.
    dest: /tmp/nagios-plugins.tar.gz
```

- This downloads the Nagios plugins needed for monitoring different services.

Extract Nagios Plugins

```
- name: Extract Nagios Plugins
  ansible.builtin.unarchive:
    src: /tmp/nagios-plugins.tar.gz
    dest: /tmp
    remote_src: yes
```

- This extracts the downloaded plugin files so they can be built and installed.

Build and install Nagios Plugins

```
- name: Install Nagios Plugins
  ansible.builtin.command: make install
  args:
    chdir: /tmp/nagios-plugins-2.4.11
```

- This builds the plugins and installs them so Nagios can use them for monitoring.

Enable and start Nagios

```
- name: Enable and start Nagios
  ansible.builtin.service:
    name: nagios
    state: started
    enabled: yes
```

- This enables Nagios to start automatically and start the service on the servers.

Enable and start Ubuntu Apache

```
- name: Enable and start Ubuntu Apache
  ansible.builtin.service:
    name: apache2
    state: started
    enabled: yes
  when: ansible_distribution == "Ubuntu"
```

- This enables and starts Apache on Ubuntu so the Nagios web interface can be accessed.

Enable and start CentOS Apache

```
- name: Enable and start CentOS Apache
  ansible.builtin.service:
    name: httpd
    state: started
    enabled: yes
  when: ansible_distribution == "CentOS"
```

- Same thing but for CentOS.

3. Show an output of the installed Nagios for both Ubuntu and CentOS.

Workstation

```
momo@workstation:~/CPE212_MOMO$ systemctl status nagios
● nagios.service - Nagios Core 4.5.14
     Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-09-17 01:25:03 UTC; 4min 7s ago
       Docs: https://www.nagios.org/documentation
    Process: 38511 ExecStartPre=/usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nag
    Process: 38512 ExecStart=/usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios
   Main PID: 38514 (nagios)
      Tasks: 8 (limit: 3459)
     Memory: 7.3M (peak: 8.7M)
        CPU: 355ms
```

Server1

```
momo@server1:~$ systemctl status nagios
● nagios.service - Nagios Core 4.5.14
     Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-09-17 01:25:03 UTC; 1min 47s ago
       Docs: https://www.nagios.org/documentation
    Process: 33672 ExecStartPre=/usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/n
    Process: 33673 ExecStart=/usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagi
   Main PID: 33675 (nagios)
      Tasks: 8 (limit: 4596)
     Memory: 7.1M (peak: 8.6M)
        CPU: 425ms
     CGroup: /system.slice/nagios.service
```

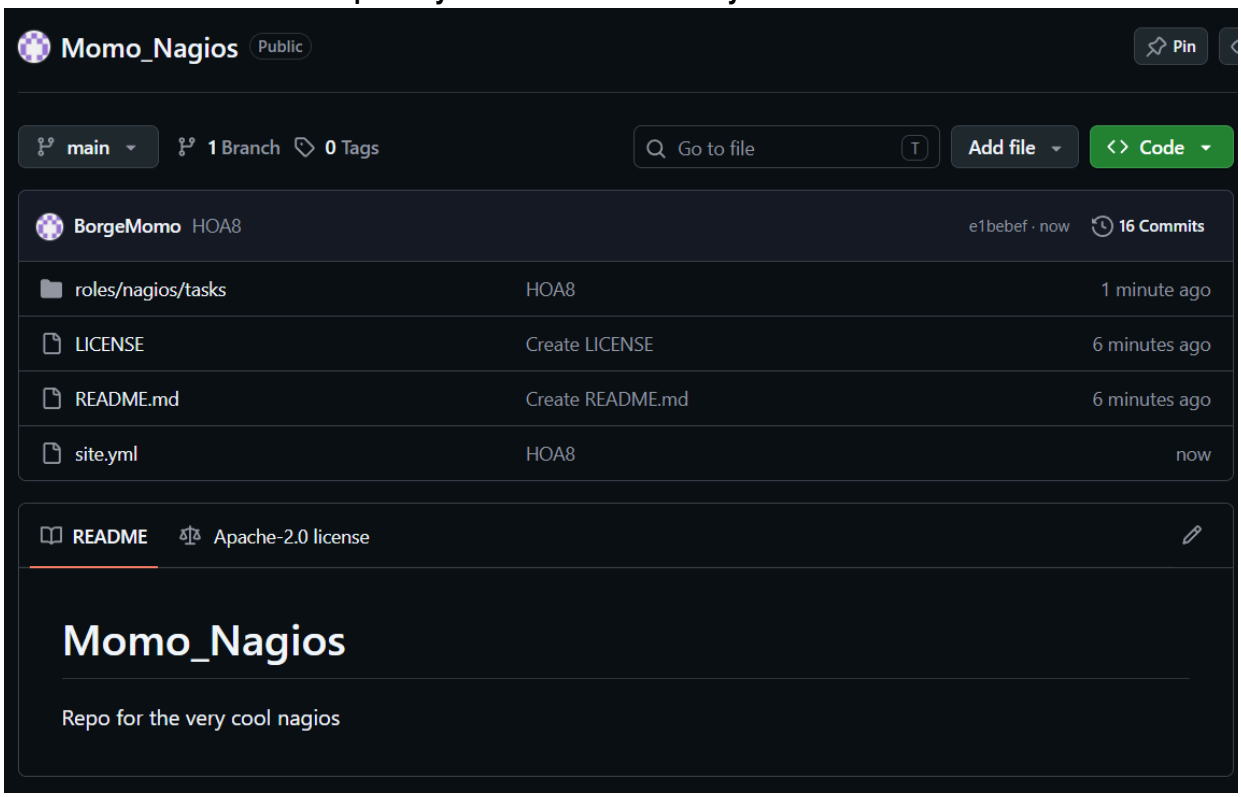
Server 2

```
momo@server2:~$ systemctl status nagios
● nagios.service - Nagios Core 4.5.14
     Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-09-17 01:25:04 UTC; 2min 44s ago
       Docs: https://www.nagios.org/documentation
    Process: 33967 ExecStartPre=/usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/na
    Process: 33968 ExecStart=/usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagio
   Main PID: 33970 (nagios)
      Tasks: 8 (limit: 4596)
     Memory: 7.2M (peak: 8.7M)
        CPU: 532ms
     CGroup: /system.slice/nagios.service
```

CentOS

```
[momo@centos ~]$ systemctl status nagios
● nagios.service - Nagios Core 4.5.14
     Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: disabled)
     Active: active (running) since Thu 2026-09-17 09:25:04 PST; 3min 22s ago
       Docs: https://www.nagios.org/documentation
   Main PID: 79721 (nagios)
      Tasks: 8 (limit: 22978)
     Memory: 29.9M (peak: 29.9M)
        CPU: 443ms
     CGroup: /system.slice/nagios.service
```

4. Make sure to create a new repository in GitHub for this activity.



https://github.com/BorgeMomo/Momo_Nagios

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What are the benefits of having an availability monitoring tool?
   - Having an availability monitoring tool can help us check if servers and services are working properly and it can also help detect problems early so they can be fixed before they cause bigger issues.

Conclusions/Learnings

In conclusion, I was able to install Nagios on both Ubuntu and CentOS using Ansible and also used roles to organize the playbook and make the installation easier to manage. The activity showed how Nagios can be used to monitor the availability of servers and services. Overall, this activity was pretty confusing at first but relying on the seatwork and using it as a guide made it easier for me to install Nagios.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Google AI**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Understanding of Concepts**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used the tool as a guide on some commands but the answers were on my own.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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